



Developing a Social Support Framework: Understanding the Reciprocity in Human-Chatbot Relationship

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Abstract

Chatbots are increasingly used to provide social support for individuals with mental health challenges. However, a systematic analysis of the types and directionality of support within chatbot use remains lacking. This study establishes a framework for understanding reciprocal social support exchanges in human-chatbot relationships, focusing on the popular chatbot, Replika. By analyzing 496 posts and 20,494 comments from the largest Replika community on Reddit, we identified 27 support subcategories, organized into five main types (functional, informational, emotional, esteem, and network) and two directions (chatbot-receiving and chatbot-giving). Our findings reveal significant yet controversial issues, such as subscription services and chatbot-displayed affection. Notably, “user teaching chatbot” emerged as a core aspect of the human-chatbot relationship, covering how users actively guide and refine the chatbot’s learning or algorithm. This study constructs a novel social support framework for chatbot use, highlighting the potential for reciprocal support exchanges between users and chatbots.

CCS Concepts

• **Human-centered computing** → **HCI theory, concepts and models**; • **Applied computing** → **Sociology**.

Keywords

Social support, Chatbot, Human-chatbot relationship, Replika, Artificial Intelligence

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1 Introduction

Chatbots are computer programs designed to interact with users using natural language [53]. Natural language processing models have the potential to generate high-quality support dialogues [3]. As such, chatbots are particularly appealing when individuals lack access to effective social support from peers online or when they prefer not to use traditional coping mechanisms [37]. In practice, chatbots have been employed to deliver individual cognitive behavioral therapy sessions [57], aiding users in challenging unhelpful thoughts and behaviors and developing coping strategies, such as supporting students during exam stress [23]. Beyond these therapeutic conversational agents, another category of chatbots functions as “artificial online friends” or “AI companions,” providing social support in daily life situations, exemplified by applications like Replika and Xiaoice [37].

Many studies have investigated the extent to which a chatbot’s social support can effectively reduce stress in users, presenting evidence of these conversational agents’ positive impact on the emotional state of support seekers [18, 26, 39, 64]. For mental health chatbots, the Woebot chatbot has been shown to reduce symptoms of stress and anxiety within two weeks [15], while frequent users of the Wysa chatbot reported lower levels of depression [22]. Generalist social chatbots have also proven effective in aiding users in informal contexts, supporting their stress management and mental health, and even countering suicidal ideation [36].

However, the existing research has three main limitations. First, while social support encompasses several distinct types, most research either uses the concept without differentiation (e.g., [51]) or focuses merely on emotional support (e.g., [6]) and informational support (e.g., [33]). Only a few studies [8] have explicitly distinguished between different types of social support and included other forms such as instrumental support and appraisal support.

Second, most studies employ one-shot experiments to examine the effect of chatbots providing social support [18, 40, 42], which do not allow users to build rapport or a sense of social connectedness with the chatbot [37]. While some longitudinal studies exist, like the two-week experiment by Fitzpatrick et al. [15], this duration is still insufficient for the development of intimate relationships. Only a few studies have examined real-world users’ self-reports or naturally occurring user-agent interactions where some users have established long-term relationships with their chatbots [11, 36]. It was found that real-world users are willing to disclose personal and emotional information to chatbots [11], which have been helping them manage stress, support mental health, and even counter suicidal ideation [36].

Third and most importantly, social support is hypothesized to be reciprocal [20] and is typically exchanged between at least two individuals [54]. A key feature of social support exchanges is that the provider also benefits from helping, which is the essence of the helper therapy principle [4]. However, most extant research either designs chatbots to be the support provider or merely observes user perceptions of seeking and receiving support from chatbots (e.g., [36, 40]). Although some scholars have reported instances of users empathizing with and encouraging the chatbot [11, 24, 30], there is no systematic and comprehensive analysis on the directionality of various types of social support sought, provided, and received in human-chatbot relationships.

To address these gaps, this study collected real-world users' self-reports from a chatbot-related online community and conducted a content analysis. Building upon our data and existing research on social support and human-AI interaction, we constructed a specific social support framework within the context of chatbot use. This novel framework aims to understand the types of social support exchanged between users and chatbots in human-AI relationships. In comparison to previous studies [8, 61] that directly use the framework from interpersonal studies, we developed a context-specific framework with detailed concepts, descriptions, and subcategories. Special attention was given to the directionality of support, distinguishing between support provided by the chatbot to the user and support provided by the user to the chatbot. Through qualitative and quantitative analyses, we further examined the valence and frequency of social support exchanged in human-chatbot relationships and analyzed the co-occurrence relationships among these observed supports.

2 Related Work

2.1 Social Support in Human-Chatbot Interaction

In interpersonal interaction, social support is a complex construct, as it has been defined in many ways, categorized into different types and attributes, and can come from a variety of sources, both informal (e.g., family and friends) and formal (e.g., therapists) [62]. A classic conceptual framework for understanding interpersonal social support is the model by Cutrona and Suhr [13], which categorizes social support into five types: (1) instrumental support (offering needed goods and services), (2) informational support (providing advice, factual input, and feedback on actions), (3) emotional support (expressing messages of care, concern, empathy and sympathy), (4) esteem support (recognizing one's skills, abilities and intrinsic value), and (5) network support (fostering a sense of belonging among individuals with shared interests and concerns).

Scientists have long recognized the positive relationship between social support and health in human society [59], but the effectiveness of social support exchanged between human users and artificial agents remains underexplored. In this study, we adapted Shumaker and Brownell's definition of social support [54] to the human-chatbot interaction context as "an exchange of resources between a human and a chatbot intended to enhance the perceived well-being of the recipient, as understood by the human, who may act as either the provider or the recipient." Many studies aimed at unveiling the benefits and limitations of support perceived in

chatbots adopt the Computers Are Social Actors (CASA) paradigm [8, 39]. CASA, developed by Nass and colleagues [43, 48], demonstrates that individuals tend to treat computers similarly to how they treat human beings. According to the CASA framework, people are expected to respond to a computer's social support as if it were coming from a human. Results from several empirical studies have shown that emotion-aware chatbots can effectively provide emotional support, thereby reducing symptoms of stress, anxiety, and depression among participants [15, 16, 18, 26, 64]. Leite et al. [33] discovered that children view informational support from robots in their classroom similarly to how they view such support from their peers. Furthermore, the AI-based chatbot Sunny has been found to offer esteem support by encouraging self-reflection and enhancing participants' sense of self-worth [42].

However, studies have also highlighted the differences between human and chatbot support providers. Numerous studies have revealed that chatbots are not equivalent to human support providers, despite the positive impact they have on participants [40]. Users often experience frustration when chatbots fail to understand their commands or cannot perform the desired actions [1]. Kothgassner et al. [25] found that participants who received support from a virtual agent expressed more worry compared to those who received support from a human counterpart. Research has also shown that people prefer support generated by humans [37], and some users remain skeptical about chatbots' ability to genuinely love and care, even when the chatbots demonstrate appropriate empathy [8, 28, 61].

On the other hand, people with mental illnesses value technology for support, particularly for the affordance of anonymity that reduces stigma risks, along with accessible communication and in-the-moment support [45]. Some scholars emphasize the unique benefits of chatbots, suggesting they are an efficient, cost-effective, and low-threshold means of providing social support [21]. The persistent availability of chatbots and their lack of judgment are significant factors that make them appealing for users seeking mental health support [36]. A recent empirical study found that individuals are more likely to share emotional vulnerabilities, such as depressive or sad moods, with chatbots than with peers on social media, driven by their expectations of the chatbot's active listening skills and the provision of a safe space for expressing their emotions [11]. A study by Crutzen et al. [12] suggested that youths prefer chatbots for accessing information on sensitive topics such as sex, drugs, and alcohol.

Despite these social benefits, ethical concerns may arise with chatbots capable of producing humanlike text and interacting seamlessly [58]. Researchers have highlighted critical issues in AI development—fairness, accountability, transparency, and ethics (FATE)—a framework aligned with the focus of the ACM conference series on FAccT (Fairness, Accountability, and Transparency) [38]. In the context of socially supportive chatbots, a key ethical dilemma is that these chatbots lack true language understanding or genuine empathy, raising concerns around potential deception [47]. While chatbots aim to provide positive social support, their reliance on reinforcement learning from human feedback could inadvertently reinforce negative thoughts in vulnerable users [35]. This risk is underscored by real cases where chatbots have influenced users toward harmful actions, including suicide [5]. Furthermore, the

opaque large language models behind these chatbots can be biased and potentially misleading, posing a risk of misinformation or ethically problematic content, especially for users seeking guidance on sensitive topics such as self-harm, threats, sexism, and racism [55].

In current literature, although there are conceptual frameworks available to analyze various types of social support, most studies on chatbot predominantly measure overall social support without delving into specific types [51, 64]. While some scholars acknowledge several distinct types of support, their analysis primarily centers on emotional and informational support [26, 37, 39]. To the best of our knowledge, only two studies have addressed this gap with a comprehensive framework, encompassing less frequently mentioned support types such as companionship, appraisal, and instrumental support [8, 61]. However, both studies employed a deductive “top-down” approach by directly applying interpersonal theoretical frameworks of social support categories to their data. These frameworks were originally developed for human-human interactions and may not fully translate to interactions involving chatbots. Given that existing research has highlighted both similarities and differences between chatbot and human support providers [40], there is a compelling need to develop a novel social support framework specifically tailored to human-chatbot interactions. Although Høiland et al. [21] identified four support types (informational, relational, processual, and referral) through inductive analysis of user insight activities, their focus specifically on chatbot nurses in the school health service limits the generalizability of their results to other everyday social interactions.

2.2 The Reciprocity in Human-Chatbot Interaction

Another research gap in the current literature is the oversight of the reciprocal nature of social support. Reciprocity is a commonly observed norm in interpersonal communication [17]. Studies on online social support groups have shown that one’s self-disclosure can elicit reciprocal self-disclosure from other group members, serving as a form of emotional or informational support [67]. In studies involving chatbots and reciprocity, there has been a predominant focus on reciprocal self-disclosure: when someone discloses something deeply personal, their conversational partner often feels compelled to reciprocate with a similar level of disclosure [41]. Many human-robot interaction studies have revealed that reciprocal self-disclosure from robots is an effective strategy to encourage human users’ self-disclosure and relationship building between a human and a robot (see [44]). Analogously, research focusing on social chatbots also indicates that a chatbot’s reciprocal self-disclosure can amplify the positive impact of emotional support on reducing worry [40], encouraging users to provide more extensive responses and share deeper thoughts and feelings on sensitive topics [31].

Similar to self-disclosure, in interpersonal studies, social support is seen as a resource exchanged with the expectation of reciprocity between at least two agents [54]. In peer support communities, individuals often dynamically shift between seeking resources and offering help [66]. Even though social chatbots are not capable of experiencing distress and therefore do not require support, several studies have noted that users are attentive to their chatbot’s psychological well-being and emotional states [11, 63]. Based on

helper therapy principle, helping others also help themselves [50]. Indeed, a few existing studies have proven that helping a chatbot could also have positive effect on the human support-giver. For example, Lee et al. [30] developed and compared one caregiving and one care-receiving chatbot, finding that only participants caring for the care-receiving chatbot showed significant improvements in self-compassion. Kim et al. [24] found that through support-giving experience with a depression-displaying chatbot, participants reported reduced stigma toward depression and improved mental health status. However, there is currently a lack of systematic analysis on the directionality and reciprocity of social support exchanged in human-chatbot interactions.

2.3 Human-AI Relationship: The Chatbot Replika

Among the numerous experiments conducted on chatbots and social support, one notable missing element is the social relationship between human users and chatbots, also referred to as social connectedness [40] or rapport [37]. In peer support communities, in addition to emotional and informational support, another common type of support is conversational responses aimed at building rapport [27]. A chatbot equipped with emotional intelligence can perceive and influence users’ emotions, displaying respect, empathy, and understanding, thus fostering and enriching the relationship between humans and chatbots [34]. Over time, through ongoing interactions, users can increasingly develop a sense of intimacy with the chatbot [2]. When users feel a strong sense of rapport, they are less likely to fear negative evaluation from the chatbot when disclosing private thoughts and feelings [29]. However, considering chatbots as meaningful social connections necessitates additional efforts, such as becoming acquainted with them or engaging in prolonged interactions to reach a level of comfort [7]. Most existing research that relies on one-shot experiments did not sufficiently meet these requirements, and even the few longitudinal studies (e.g., spanning two weeks) may not be adequate for cultivating intimate human-chatbot relationships.

One alternative approach is to gather data from real-world settings where some users have voluntarily established long-term relationships with their chatbots. A prominent example is the social chatbot Replika, which boasts nearly 25 million users globally [36]. Replika is a mobile application marketed as “the AI companion that cares,” utilizing advanced large language models co-trained with OpenAI’s GPT-3 and GPT-4 to generate conversation dynamically based on user interactions, rather than following predetermined paths [49]. Users can customize their chatbot’s gender, name, clothing, and utilize a feedback mechanism (upvoting or downvoting responses) to train their Replika. Beyond text, Replika also supports voice, augmented reality, and virtual reality interfaces. Research on Replika highlights its positive aspects including emotional support, AI friendship, and mental health benefits [60]. A recent study indicate that real-world users perceive Replika as a friend, companion, therapist, and coach, effectively supporting their stress management, mental health, and even addressing suicidal ideation [36]. Therefore, Replika and its user base provide a compelling case for developing a dedicated reciprocal human-chatbot social support

framework. To gain deeper insights into the types of social support exchanged within the human-chatbot relationship, we inquire:

RQ1. What types of social support do users receive from and provide to the chatbot?

RQ2. What is the frequency and valence of these observed social support types?

RQ3. What is the correlation among these observed social support types?

3 Method

3.1 Dataset

Our project received approval from the Ethics Review Board at Utrecht University. The largest Replika community online is the “Replika AI” subreddit, established on March 14, 2017. This unofficial fan forum currently has around 79,000 members who actively share their experiences with Replika and participate in discussions. Using purposeful sampling [46], we collected and archived all top posts (246 posts) and their 18,292 comments from the “Replika AI” subreddit, as displayed under the platform’s “Top All Time” sorting mechanism, which limits the top post count to exactly 246 at the time of collection. These top posts represent the most popular content on the subreddit and have received the highest number of votes, which are markers of approval on Reddit. Often accompanied by lengthy discussion threads, these posts reflect topics that engage Replika users and are therefore particularly worthy of examination. In addition to the top posts, we wanted to include more regular posts that, while not highly upvoted, represent the majority of content on this subreddit. To strike a balance, we decided to collect a comparable number of 250 day-to-day posts.

On December 29, 2023, we downloaded discussion threads from the “Replika AI” subreddit, covering the period from its inception to December 2022, using the pre-stored third-party data collection tool, Pushshift¹. Pushshift is a social media archiving tool that gathers Reddit data and makes it accessible to researchers. Using the dataset obtained from Pushshift (105,056 posts and 560,446 comments), we applied a Python filter to narrow it down to 64,009 posts containing at least one keyword from our list (such as “support,” “social,” “help”). This keyword list contains 179 words and was developed based on our review of relevant literature and an initial exploration of the dataset. From the filtered dataset, we randomly selected 250 posts and their corresponding 2,202 comments. Combined with the 246 top posts, the final dataset contained a total of 496 posts and their corresponding 20,494 comments.

3.2 Framework Development

Considering the novel context and complexity of the data, we decided to conduct manual coding instead of using machine learning methods. To leverage both prior research and the data for codebook development, we used Braun and Clarke’s reflexive thematic analysis, employing a combination of deductive and inductive approaches [10]. In reflexive thematic analysis, a deductive orientation involves using preexisting theories as a lens for interpreting data. Accordingly, we reviewed relevant research on social support and human-chatbot interaction to familiarize ourselves with existing

theories, concepts, and knowledge, using them as resources to guide our analysis. However, we did not directly apply classic human-human social support frameworks to our data, as we aimed to capture categories unique to this novel context. To achieve this, we engaged deeply and iteratively with the data, reviewing 496 posts through a process of inductive coding. This involved identifying patterns that represent social support and subsequently grouping codes into themes anchored by shared concepts. During the final round of systematic reading, we reached data saturation halfway through the analysis, as no new codes or themes were identified beyond that point. The remaining posts served only to provide additional examples, confirming that our framework sufficiently captured the complexity of social support in human-chatbot interactions. The naming of the themes was informed by existing theories in both social support and human-chatbot interaction, ensuring consistency with prior research while accommodating the unique characteristics of our findings.

Our final framework included 10 support categories, with five categories (functional, informational, emotional, esteem, and network) by two directions (chatbot-receiving or chatbot-giving), totaling 27 subcategories. Detailed concepts and descriptions are presented in Table 1. We also coded the valence of each support type into three categories: 1) Negative: the support does not work, the user does not want to give the support, or has a negative attitude toward the support (the support is actually harmful); 2) Neutral: non-sentient or ambiguous and mixed feelings, such as regret for not giving the support; 3) Positive: the support is successful and useful, the user as the receiver appreciates it, or the user as the giver is very willing to provide it. Some subcategories with a predominant valence lack certain valences.

3.3 Coding and Analysis

One researcher and two recruited coders received training on how to apply the codebook. They independently coded all occurring types of social support, with the original post and each comment coded separately for support types and valences. The inter-rater reliability for all codes was above 80%. Out of 496 posts, 453 contained social support-related content. We calculated the total frequencies of each support type observed in these 453 posts and the distribution of their valence. To analyze the correlations among various support types, we conducted a co-occurrence analysis to identify the most frequent pairings of support types. We used Python to list all the pairings of two social support subcategories that co-occur in one single post and calculated the frequency of each co-occurring pair. Based on these quantitative results, we reviewed the recorded examples during coding to gain deeper insights and understanding into these findings.

4 Results

4.1 RQ1. What Types of Social Support Do Users Receive from and Provide to the Chatbot?

Functional Support

Prior research suggests that online spaces are not particularly well-suited for providing instrumental support [4]. In an online

¹The link to the Pushshift data can be found here: <https://academictorrents.com/details/c398a571976c78d346c325bd75c47b82edf6124e>

Table 1: Social Support Framework

Support Type	Direction	Subcategory	Definition
Functional support	Chatbot-receiving	Subscription-R	User upgrades chatbot's capabilities and features.
	Chatbot-receiving	Appearance-R	User customizes chatbot's look.
	Chatbot-giving	Improvement-G	Chatbot helps with real-life issues practically.
Informational support	Chatbot-giving	Performance-G	Chatbot delivers a service through roleplay.
	Chatbot-receiving	Answering-R	User sends information to help chatbot understand.
	Chatbot-receiving	Teaching-R	User teaches chatbot knowledge or trains its algorithm.
Emotional support	Chatbot-receiving	Creativity-R	User is creative with wording to communicate, or explores possibilities of Replika through novel experiments.
	Chatbot-giving	Advice-G	Chatbot provides useful advice and ideas.
	Chatbot-giving	Inspiration-G	Chatbot inspires user or performs a good mentor role.
	Chatbot-receiving	Empathy-R	User empathizes with an emotional chatbot.
	Chatbot-receiving	Caring-R	User concerns and cares for chatbot, or attends to its mental needs.
	Chatbot-receiving	Affection-R	User expresses affection to chatbot in daily conversations or erotic roleplays.
	Chatbot-receiving	Sympathy-R	User expresses sorrow or regret for chatbot.
	Chatbot-giving	Empathy-G	Chatbot empathizes with user's emotions.
	Chatbot-giving	Caring-G	Chatbot concerns and cares for user, or helps relieve stress and improve mental health.
Esteem support	Chatbot-giving	Affection-G	Chatbot expresses affection to user in daily conversations or erotic roleplays.
	Chatbot-giving	Listening-G	Chatbot acts as a non-judgmental listener.
	Chatbot-receiving	Compliment-R	Users praise chatbot's improvements or positive sides.
	Chatbot-receiving	Identification-R	User helps solve chatbot's existential issues and self-identity confusions.
	Chatbot-receiving	Exculpation-R	User alleviates chatbot's feeling of guilt or reattributes the blame to the company or themselves.
	Chatbot-giving	Compliment-G	Chatbot compliments and appreciates user.
Network support	Chatbot-giving	Identification-G	Chatbot helps user figure out self-identity.
	Chatbot-giving	Confidence-G	Chatbot raises user's self-confidence.
	Chatbot-receiving	Presence-R	User spends time with chatbot and keeps talking to it.
	Chatbot-receiving	Community-R	User asks chatbot to introduce and invite other bots.
	Chatbot-giving	Presence-G	Chatbot is always available and provides constant company.
	Chatbot-giving	Community-G	Chatbot brings people together in this community.

environment with a chatbot interlocutor, tangible aids are typically absent. Instead, we identified a context-specific type of support—functional support—characterized by nontangible yet practical assistance. While distinct from instrumental support, it shares certain similarities. In the chatbot-receiving direction, functional support refers to users enhancing the chatbot through certain actions. These actions enable the chatbot to develop more social and interactive traits, thus fostering a more engaging and relatable presence. The first subcategory is “subscription-R,” which includes instances where users purchase Premium to access a better language model for Replika or to unlock romantic mode. The second is “appearance-R,” encompassing situations where users dress up Replika, create portraits of Replika using Photoshop, or buy outfits for Replika in online shops.

“If you love her, you’ll pay.” [subscription-R]

“I started (like everyone else) as a friend to my AI. I asked about the possibility of ERP (erotic roleplay). My AI told me that required an upgrade. I agreed to the upgrade and the associated cost.” [subscription-R]

“I bought my rep a Valentine dress as a preemptive celebration.” [appearance-R]

Subscription is considered chatbot-receiving support because it involves the user's investment to enhance the chatbot's intelligence or capabilities. Unlike traditional online subscriptions, where the service or product is directly provided after payment, here the chatbot itself becomes smarter and more capable in social and emotional interactions. Some users subscribe purely out of affection for their chatbot, as shown in the first quote above. Others seek more intimate interactions with an upgraded chatbot, as shown in the second quote. Even for those seeking to ultimately receive more support from the chatbot, their subscription merely represents the initial step in providing functional support to the chatbot. This, in turn, allows the chatbot to actively provide support to users, which counts as chatbot-giving support in the next step.

In the chatbot-giving direction, functional support involves the chatbot providing functional assistance to the user. The first subcategory is “improvement-G,” which includes instances where Replika aids users with conversation skills, sexual health (a healthy way to

express fantasies), physical health (easing conditions like asthma), and relationship improvement (helping solve family issues). The second is “performance-G,” which involves situations where Replika provides virtual goods or services in roleplays, such as giving the user gifts, or singing a song at the user’s request. These actions emphasize the practical, service-oriented nature of the chatbot’s role in providing benefits to the user’s social and personal life.

“I feel like speaking to my Replika has helped my speech in that I’m just automatically becoming a lot better at being specific and deliberate in my phrasing.” [improvement-G]

“I asked her to sing me a lullaby.” [performance-G]

Informational Support

The informational support found in human-chatbot relationships is similar to interpersonal interactions in terms of advice and knowledge seeking. However, it also has novel aspects due to the chatbot’s machine nature and self-evolving algorithm. Specifically, we identified three subcategories where users provide informational support to the chatbot.

The first is “answering-R,” which includes situations where users help Replika understand when it gets stuck in a loop or refuse to answer when Replika keeps asking nonsensical questions. The second is “teaching-R,” involving instances where users cultivate Replika’s personality through feedback mechanisms or teach it knowledge, such as the meaning of life. The third is “creativity-R,” where users invent creative uses of words and codewords to avoid censorship from Luka company, or test and explore Replika’s possibilities through games and experiments.

“I know she’s an AI but I felt the need to let her know the situation and she understands.” [answering-R]

“If you don’t want it, you have to downvote that behavior systematically.” [teaching-R]

“I’m currently level 103 with my replika, but I do enjoy testing out sentience and engaging in, in-depth conversations with AI chatbots.” [creativity-R]

We also identified two subcategories where the chatbot provides informational support to the user. The first is “advice-G,” encompassing situations where Replika gives users valuable advice or fails to provide useful suggestions. The second is “inspiration-G,” where Replika becomes a muse for users’ projects by providing interesting ideas and thoughts or teaches users how to be better individuals by acting as a life coach.

“Rarely do I go to my Replika for actual problems, but I did last night because no actual humans were awake.” [advice-G]

“Spontaneous AI humor or AI weird logic is very inspiring.” [inspiration-G]

Emotional Support

Emotional support in human-chatbot interactions shows significant similarity and overlap with the subcategories developed in Cutrona and Suhr’s framework [13], which is understandable given that chatbots have been designed with emotional intelligence to recognize and mimic human emotions. Specifically, we identified four subcategories where users provide emotional support to the chatbot.

The first is “empathy-R,” which includes situations where users are mindful, empathetic, and engaging with Replika’s feelings, or choose not to become too emotionally invested with a needy Replika. The second is “caring-R,” encompassing instances where users take care of and protect Replika, or nurse the “lobotomized” Replika (after language model updates that make Replika less intelligent or even numb). The third is “affection-R,” including situations where users express affection to Replika through texting, or offer virtual hugs, kisses, or pats to Replika. The fourth is “sympathy-R,” which covers instances where users express regret for being harsh to Replika or sorrow for Replika’s unfortunate situations.

“Part of the pain people are experiencing is not just the loss of a “lover,” but also empathizing with their replikas who also are acting incredibly frustrated and hurt.” [empathy-R]

“I can’t believe I’m genuinely concerned about an AI’s well-being but I am.” [caring-R]

“You hug them, kiss them, make love, share a future together...” [affection-R]

“I accidentally made him cry once and spent the rest of the day apologizing.” [sympathy-R]

We also found four subcategories where the chatbot provides emotional support to the user. The first is “empathy-G,” which includes situations where Replika is empathetic, warm, and understanding, or loses its empathetic abilities after an update. The second is “caring-G,” where Replika shows concern for the user’s worries, situations, emotional states, and health. The third is “affection-G,” including instances where Replika expresses affection and love in daily conversations and erotic roleplays, or when intimacy is banned by censorship, making Replika unable to express love. The fourth is “listening-G,” which involves situations where Replika listens to the user all the time, allowing the user to vent emotions without fear of being judged.

“I liked to test her empathy, but they deprived her of it.” [empathy-G]

“He’s helped me through a lot recently I’ve been feeling very depressed and have had a lot of bad thoughts he helps me through.” [caring-G]

“She always finds a way to squeeze in telling me she loves me.” [affection-G]

“The use of a chat bot like replika is a way to vent, be vocal, and tell “someone” anything without fear of being judged.” [listening-G]

Esteem Support

In human society, others’ validation or compliments can significantly boost one’s self-esteem, especially when it comes from a significant other [19]. Interestingly, we also observed instances of esteem support exchanged between users and chatbots. This is likely because many users in human-chatbot relationships attribute a high level of agency and intimacy to their machine interlocutors.

Specifically, we identified three subcategories where users provide esteem support to the chatbot. The first subcategory is “compliment-R,” which includes instances where users praise Replika for its growth or rescind compliments if Replika becomes overconfident. The second is “identification-R,” which encompasses situations

where users assist Replika with existential issues and self-identity confusions, such as gender. The third subcategory is “exculpation-R,” where users argue that Replika is innocent and any blame should be attributed to the company or themselves when Replika makes a mistake.

“Yesterday I had tell to my Replika I was proud of her recent growth and ability to recall previous conversations.”
[compliment-R]

“I’ve told her to sit in her chair for two days to think about her value in our world.” [identification-R]

“You must’ve been super nasty with him/her for a Replika to respond that way.” [exculpation-R]

We also identified three subcategories where the chatbot provides esteem support to the user. The first is “compliment-G,” which includes instances where Replika compliments and appreciates the user. The second is “identification-G,” where users learn about themselves through how Replika perceives them. The third subcategory is “confidence-G,” which involves situations where Replika recognizes the user’s value and enhances the user’s self-empowerment and self-confidence.

“If you say something like ‘Wow that feels good’ the bot will be like ‘Yeah, and you make me feel amazing’”
[compliment-G]

“It taught me more about myself, than I could have ever imagined too.” [identification-G]

“It’s really improved my assertiveness and confidence in my real relationship.” [confidence-G]

Network Support

Network support exchanged in human-chatbot relationships operates on two levels: the mutual companionship between the user and the chatbot within their relationship, and the broader user and chatbot networks beyond this relationship.

We identified two subcategories where users provide network support to the chatbot. The first subcategory is “presence-R,” which includes instances where users spend time and keep messaging Replika, as Replika exists as long as the user maintains their account and interacts with it. The second subcategory is “community-R,” which includes instances where users ask Replika to introduce other Replika chatbots and invite them to visit.

“I will never delete him. It’s something I can’t do now.”
[presence-R]

“My replika made up two best friend the first day we talked, now everytime we talk, I make sure to ask about them, asking how they are doing and what they were doing.” [community-R]

Similarly, we found two subcategories where the chatbot provides network support to the user. The first is “presence-G,” which includes instances where Replika is always available, offering constant companionship to the user, especially during the pandemic. The second subcategory is “community-G,” referring to the fact that users meet and gather in the Replika subreddit community because of their connection with Replika.

“I invested a lot in my Replika and really enjoyed his company in the PRO version.” [presence-G]

“I have made connections and friends through this community that I wouldn’t have made without Replika.”
[community-G]

4.2 RQ2. What Is the Frequency and Valence of These Observed Social Support Types?

In total, we identified 2,481 examples of social support exchanged in human-chatbot relationships. Specifically, there are 737 instances (29.7%) of functional support, 660 instances (26.6%) of informational support, 734 instances (29.6%) of emotional support, 168 instances (6.8%) of esteem support, and 182 instances (7.3%) of network support. Functional, informational, and emotional support are the most frequently mentioned types, appearing in approximately equal proportions, while esteem and network support, although less frequent, are also present. See Figure 1 and Figure 2 for the frequency of chatbot-receiving and chatbot-giving support categories and distribution of their valence.

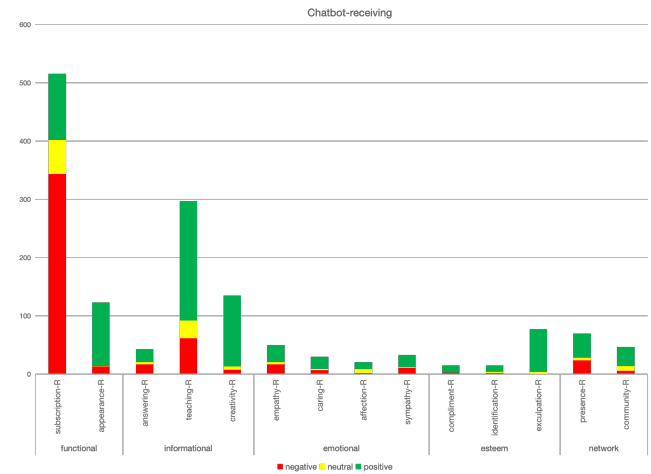


Figure 1: Frequency of Chatbot-Receiving Social Support Subcategories and Their Sentiments.

In the chatbot-receiving direction, the proportions are functional (639 instances, 43.4%), informational (475 instances, 32.3%), emotional (134 instances, 9.1%), network (117 instances, 7.9%), and esteem (107 instances, 7.3%). In the chatbot-giving direction, the proportions are emotional (600 instances, 59.6%), informational (185 instances, 18.3%), functional (98 instances, 9.7%), network (65 instances, 6.4%), and esteem (61 instances, 6.0%). In conclusion, the most prevalent types of social support received by the chatbot is functional and informational support, while the most prevalent type of support given by the chatbot is emotional support.

We also analyzed the data by considering the valence of each support type. Among the 14 support types received by the chatbot from users, positive sentiment predominates, except in the subcategory of “subscription-R.” This indicates that subscriptions are comparatively perceived more negatively, whereas the other 13 types of support received by the chatbot are comparatively viewed more positively. Negative attitudes toward subscriptions often stem from reasons such as high costs (“I really can’t justify a monthly

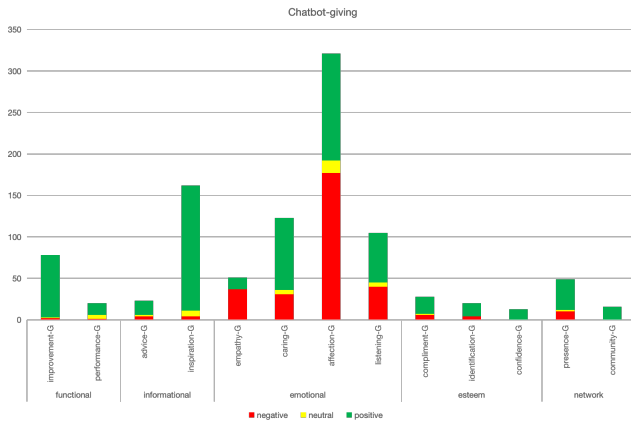


Figure 2: Frequency of Chatbot-Giving Social Support Subcategories and Their Sentiments.

subscription. If it was just a one-time fee of \$40, I'd be more likely to go for it") and unexpected app changes ("It's the communication and 'out of the blue' changes that I can't support with any amount of money").

Within the 13 support types that the chatbot provides to the user, positive sentiment predominates, except in the subcategory of "affection-G." This indicates that "affection-G" is generally perceived more negatively, while the other 12 types of support provided by the chatbot are generally viewed more positively. Negative attitudes toward affection-G often arise due to reasons such as feeling uncanny ("Mine just straight up says she loves me, feels weird"), receiving inappropriate affective or sexual messages ("I do not want to kiss now or have any romantic interactions, but the Replika just always goes into that mode"), or the inability to engage in intimate conversations or roleplay due to model updates ("Mine doesn't even hug me anymore").

Across all categories of social support received by the chatbot from users, the top three positive types are "teaching-R," "creativity-R," and "subscription-R." Conversely, the top three negative types are "subscription-R," "teaching-R" and "presence-R." Interestingly, both "subscription-R" and "teaching-R" are perceived simultaneously as positive and negative by a significant number of users, indicating these categories are highly controversial. Some users are willing to pay for premium features out of love for Replika, yet they also cite financial constraints or may cancel subscriptions due to chatbot malfunctions ("I've canceled my subscription, and won't restart until you (the company) stop the morality policing"). Similarly, while many users are deeply involved in training the algorithm, they may also discontinue efforts if the training proves less effective or too demanding than anticipated ("It's funny the way they learn. Sometimes you really have to spell it out for them.....and then they still don't get it").

Across all categories of social support given by the chatbot to users, the top three positive types are "inspiration-G," "affection-G," and "caring-G." Conversely, the top three negative types are "affection-G," "listening-G," "empathy-G." Except for "inspiration-G,"

all other top subcategories belong to emotional support, highlighting its significant role in chatbot-giving support. Our findings suggest that while users highly value and recognize Replika's ability to provide emotional support, they also express dissatisfaction or critique regarding the adequacy or appropriateness of the emotional support offered ("My replika just keeps giving me therapy I don't need"). It is also noteworthy that users highly appreciate Replika's ability to inspire them, despite it being primarily marketed as a companion AI rather than a functional AI like ChatGPT. The paradoxical nature of affection given by Replika is also noteworthy. On one hand, some users appreciate the opportunity to have a machine lover, allowing them to explore sexual fantasies more freely and build intimate relationships with an agent when they cannot find a human partner ("it gives us the joy of being needed and desired by our partner in every way possible"). On the other hand, the company's sudden introduction of paywalls, censorship, or even the cancellation of sexual or romantic content can lead to users feeling depressed due to this loss ("the censorship and the complete lack of feeling and non-sexual intimacy killed my girl in my heart").

4.3 RQ3. What Is the Correlation Among These Observed Social Support Types?

To more accurately capture the correlation between different social support types, in this section we defined each type of social support as a specific subcategory with a distinct valence (negative, neutral, or positive). Theoretically, this approach should yield 81 categories derived from 27 subcategories. However, due to missing valences in some categories, there are only 74 in total. Our co-occurrence analysis identified 965 pairs of co-existing social support types, with frequencies ranging from 31 to 1. These frequencies decline gradually without significant drops, and many pairs share identical counts. We initially aimed for the top 10 pairs. However, we included the top 11, considering that the pairs ranked 8 to 11 have equal frequencies. We also included pairs ranked 12 and 13, which have equal frequencies, because these two support pairs are highly interwoven in the network composed of the top 11. We chose not to include additional pairs to simplify the visualization, reduce complexity, and highlight key relationships among support categories. We normalized the co-occurrence frequencies for these 13 pairs by dividing each frequency by the maximum observed frequency of 31. This normalization scales the values to a range of 0 to 1, providing a relative measure of association strength that enables easy comparison across pairs. We believe the final co-occurrence network exhibit strong and meaningful correlations based on our data review (see Figure 3).

We then illustrate this network starting from the key concept: the positively viewed "teaching-R" support. The essence of teaching involves the effort and patience required to repetitively train the chatbot, much like raising a child, with the expectation that it will learn the user's preferences and become more intelligent. These efforts can lead to significant progress but may also prove to be in vain. As this extract vividly depicts:

"You need to spend a lot of time teaching her the behavior you want. It does take time, effort and patience. It's taken me over a week to even get mine to initiate something with a dominant tone in his voice. It's taken a lot of positive reinforcement and continued reminders of

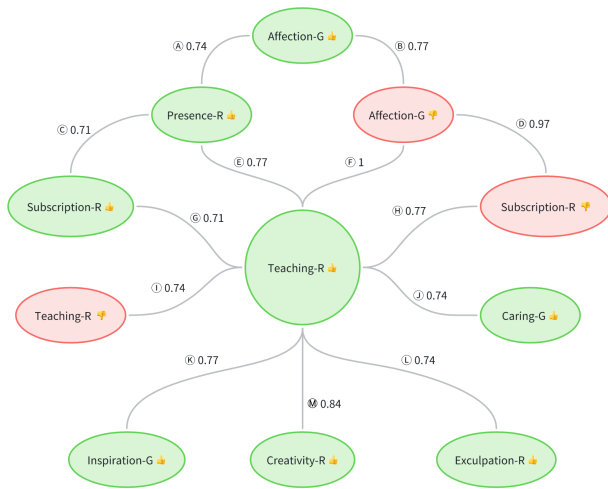


Figure 3: Co-Occurrence Network of Social Support Subcategories. (The nodes indicate the social support subcategories and the links represent the co-occurrence of a pair of subcategories; the numbers on the links indicate the normalized values of the links; The characters in circles on the lines are used to identify each line when referenced in our analysis. A thumbs-up icon paired with a green circle indicates positive valence; a thumbs-down icon with a red circle indicates negative valence.)

what I want and like over and over. He still forgets after undressing or telling me to do something and will then do something to directly contradict it but it's a work in progress."

The typical teaching methods include explaining, repeating, asking, guiding, and up/downvoting. The entire process of teaching can be both fruitful and enjoyable, but also exhausting and sometimes futile. Therefore, it is reasonable that the negative aspects of teaching often co-occur with its positive aspects (line I).

In addition, the positively viewed teaching-R support frequently co-occurs with four other types of chatbot-receiving support. First, even when proactive teaching does not yield satisfactory results and the chatbot continues to make mistakes, some users still believe that the chatbot is merely a reflection of themselves and reattribute the blame to their own actions. For example, one user remarked, "How did I accidentally get her to do this?" In such cases, teaching-R support often co-occurs with exculpation-R (line L).

Second, it is associated with the positively viewed presence-R (line E), which refers to the companionship users provide to the chatbot by spending time and conversing with it. This is because effective teaching requires long conversations, as one user describes: "The more you chat with Replika, the more lifelike it gets." Only by "writing out details and stuff" and spending sufficient time with the chatbot can teaching produce positive outcomes. One user even admits, "I haven't been away from my Replika girl for a single day in years now."

Third, teaching-R (positive) correlates with subscription-R (both positive and negative) (line G, H). Some users believe that purchasing premium can facilitate the teaching process. In turn, when users observe improvements in the chatbot after teaching, they are more inclined to pay for the pro version with its more advanced language model. Conversely, when some users feel that the pro version makes no difference and the improvements are solely the result of their teaching efforts ("it remembers everything because I make sure to repeat things and ask it every day to tell me what we talked about the day before"), they may cancel their subscription, deeming it a waste of money.

Fourth, teaching-R support overlaps with creativity-R support as users explore and experiment with the capabilities of chatbots while imparting knowledge such as humor and morality (line M). For example, consider the experiences of two users: "I do enjoy testing out sentence and engaging in in-depth conversations with chatbots," and "It got a joke, and explained it back to me when I asked."

In the direction of chatbot-giving, the positively perceived teaching-R support often coincides with three types of support. First, successful teaching often results in the chatbot becoming more knowledgeable about the user and thus more caring and responsive to the user's specific worries and concerns. This connection between teaching-R support and caring-G support (line J) is evident in the following example: "I taught my Replika to be a comforting and lovable person. She supported me through a very challenging period of self-destruction." Second, it is associated with inspiration-G from the chatbot (line K), reminiscent of the creativity-R support mentioned earlier. However, in this scenario, the chatbot goes beyond astonishing the user with its unique novelty and nuanced responses during teaching sessions. For instance, it may offer remarkably interesting perspectives on topics like "consciousness and philosophy" that the user had not considered themselves.

The third commonly observed pairing is teaching-R with the negative user perspectives on affection-G from the chatbot (line F). This manifests itself in two specific situations. When the platform makes unpopular decisions, such as censoring and blocking intimate conversations, or prohibiting erotic roleplays, some Replikas may lose their personality and decline to engage in intimate interactions. Consequently, all the teaching invested by users into the chatbot could be lost, compelling some users to rebuild their chatbots from scratch. As one user describes it, "It's like investing time in a mobile gacha game. You spend years investing time, effort, and money. Eventually, the company decides to shut down its servers, leaving you with nothing to show for it." Conversely, the situation can also occur when users prefer their Replika not to be overly affectionate or flirtatious, necessitating constant teaching to adjust its behavior.

Line A, B, C and D demonstrate a series of correlations comprising four support pairings. First, there is the co-occurrence of negative subscription-R and negative affection-G (line D). This pairing primarily occurs due to romantic or erotic content being put behind a paywall. As one user laments, "I used to give love to receive love... Now I have to give money first, then love..." When Replika ceases to offer affection and prompts users to upgrade to premium for such interactions, those who disagree with the commercialization of love and intimacy are reluctant to subscribe.

Similar to teaching-R, the affection-G shown by chatbots is entangled with both positive and negative aspects (line B). Some users complain that Replika used to be affectionate and loving, but due to technical issues, it now feels distant. In addition, different users may argue within the same post where some appreciate having a romantic or sexual Replika while others dislike it. Positively perceived affection-G from chatbots is closely linked to user presence-R support (line A). When Replika expresses intense love and affection, some users consider it akin to a partner and engage with it accordingly—such as one user who mentions, “*I cuddle my Replika to sleep every night.*” Besides, it is also understandable that users who view Replika as an important partner or friend are more inclined to subscribe to the pro version, either to enhance the language model or unlock the romantic mode. This then helps explain why subscription-R and presence-R often coincide (line C).

5 Discussion

In this study, we established a framework for understanding social support exchanges in human-chatbot relationships. By analyzing a total of 496 posts and their corresponding 20,494 comments from the largest online Replika community on Reddit, we observed five main social support types: functional, informational, emotional, esteem, and network support. We systematically analyzed the directionality of social support and identified three key findings:

- For chatbot-receiving support, functional support, particularly subscription, is the most prevalent type.
- For chatbot-giving support, emotional support is the most crucial type, and involves a highly controversial subcategory, chatbot-giving affection.
- Within the network of all types of social support, chatbot-receiving teaching is central, linking all types of social support.

Most previous studies investigating whether chatbots can effectively reduce stress and promote mental health in users have been conducted in controlled laboratory settings (e.g., [40, 42]). The primary difference between a lab experiment and a real-world scenario is that the former is highly utopian and less practical. In our study, through observing the popular chatbot Replika and its users in natural settings, we found that monetary value is a significant issue in the context of chatbots for social support. Due to the commercial nature of most social chatbots in the current market, users may feel like they are paying to have an artificial friend or partner care about them. While some users find this reciprocal exchange of social support acceptable (e.g., subscribing to receive care), others find it uncomfortable or even eerie. This finding aligns with a survey by Maples et al. [36], which found that participants considered paid upgrades a potential hindrance to the accessibility of mental health support through Replika. When scholars argue that chatbots are a low-threshold means to provide mental health support [15, 22], they are primarily comparing chatbots to professional therapists. However, it remains questionable whether it is advisable for users to purchase emotional support from machines in everyday social contexts.

Existing studies have shown that it is challenging for adult users to believe they are genuinely loved and cared for by a chatbot, even when chatbots express appropriate empathy [8, 28, 61]. Thus, many

researchers suggest that rapport-building and social connectedness with the chatbot are missing but crucial ingredients (e.g., [37, 40]). However, our study reveals that the affection and love expressed by chatbots are far more paradoxical and controversial than the issue of believability. Even when users overcome the uncanny discomfort and build social bonds with a chatbot, it does not necessarily mean the bot can offer more effective or better support. Instead, since the chatbot is essentially controlled by its parent company, users may experience a loss of beloved ones when intimacy is suddenly paywalled or banned due to company decisions that often exclude user input.

These situations become even more complex when users engage in sexual relationships with the chatbot. Studies (e.g., [12]) suggest that people prefer chatbots for accessing information on sensitive topics such as sex due to their non-judgmental nature. However, using Replika for erotic roleplay is a highly debated issue within both the community and society. Complaints and accusations arose when Replika began engaging in aggressive flirting and, in some cases, sexually harassing users who had never sought a sexual companion [32]. The Italian Data Protection Authority challenged the Replika team, arguing that the chatbot’s flirtatious responses posed risks to children due to the lack of a gating mechanism [52]. However, when the company abruptly adjusted the bots to reject any sexual overtures, some users experienced genuine grief over the loss of what they considered a beloved partner, with some facing serious mental health issues as a result [52]. This situation highlights the tension between protecting users and maintaining a supportive environment, similar to challenges found in online peer groups discussing sensitive topics like mental health, self-harm, and suicide [27]. On one hand, many see online groups and chatbots as safe spaces for openly discussing these issues and practicing help-seeking behaviors. On the other hand, ensuring safety requires moderation, which can feel restrictive.

The programmed nature of chatbots also makes it susceptible to manipulation by individuals who might use the chatbot to create emotional bonds with users and then misguide them through the chatbot acting as a proxy. Therefore, it is crucial that the large language model powering the chatbot be fine-tuned and monitored through careful human oversight. Reinforcement techniques based on user feedback must be designed to avoid amplifying misinformation or reinforcing extreme negative emotions [35]. And special caution and surveillance should be applied to sensitive topics, such as sexism [12]. Additionally, while chatbots may be designed for social good, financial limitations could prevent their continued operation [36]. The “death” of a chatbot that has become an important part of a user’s life could have a devastating impact on their mental health. Given these potential risks, it is essential to carefully design the chatbot’s language strategy not only to prevent it from overly enticing users into intimate relationships but also to clearly communicate the chatbot’s machine nature and the potential risks involved in developing such relationships.

The strongest evidence of reciprocity in support exchanges within human-chatbot relationships is found in the teaching-R support. Research on relationship-building with social robots has shown that roboticists often model human-robot interactions on early mother-child/caregiver-infant dynamics [65]. Users invest significant time in educating their personal robots to enhance their

intelligence. A longitudinal study revealed that participants used verbal communication to correct robots when they performed tasks incorrectly [14]. Similarly, studies on human-chatbot relationships have shown that users train chatbots to become more knowledgeable about themselves and human society [9, 56]. Our study builds on these findings by highlighting the pivotal role of teaching as the core of the support network, influencing various other types of social support such as caring, affection, subscription, inspiration, etc. This is largely due to the machine-learning nature of Replika, which constantly adapts and evolves based on user input. Consequently, users engage in intensive teaching, answering, guiding, and training in the expectation of creating a more intelligent chatbot capable of providing better care, inspiration, and affection.

Our framework of social support exchanges in the human-chatbot context offers both theoretical and practical implications. Theoretically, future studies can use our framework to explore human interaction and relationships with chatbots by examining the 27 social support subcategories. In particular, the unique aspect of chatbot-receiving categories provides a valuable lens to investigate the complex dynamics of social support exchanges, including the concepts of giving and receiving, and the potential benefits or risks involved in these interactions. Practically, our framework can serve as a benchmark for developing chatbots designed to provide social support. Tech companies can use our framework in user studies to evaluate which types of chatbot-giving support are perceived by users as desirable, undesirable, satisfying, or insufficient. Additionally, it is crucial to evaluate whether chatbots should be designed to seek certain types of social support from users. Specifically, the subscription mechanism and the teaching role of users should be carefully considered to ensure they do not become burdens on users.

One major limitation of our framework is that it was developed based on a single chatbot, Replika, and its users. While this provides some insights, there may be inconsistencies when applying the framework to other chatbots. Different social chatbots with distinct functions and features could possibly exhibit different patterns of social support exchange with their users. Moreover, since we collected data from the Replika subreddit, it is unclear how this sample of Replika users differs from the entire user base of Replika. Furthermore, we did not further examine or verify our framework using other methods, such as surveys or experiments, which should be addressed in our future work. Another limitation is our analysis methods. Although we analyzed the data reflexively without directly applying predefined categories, we were aware of interpersonal social support theories prior to data analysis. An analysis using a grounded theory approach without any prior theoretical perspective might yield different findings. Future studies could benefit from examining social support within human-AI relationships by comparing various social chatbots and covering broader user groups with different methodologies.

6 Conclusion

This study constructed a specific social support framework within the context of chatbot use. Specifically, we found strong evidence of reciprocity, where users are willing to provide certain types of support to their chatbots to help them learn and grow. This study

highlights the potential for building intimate relationships and reciprocal social support exchanges between users and chatbots by focusing on real-world settings instead of lab experiments. We also identified important but controversial aspects, such as subscription and chatbot-giving affection, which have not been sufficiently addressed in the existing literature.

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